

CS (Diploma) — MS Office & MS-DOS (Paper-I §8, Paper-II §1)

Elective 3: **Computer Science / Computer Engineering (Diploma)**, NPSC CTSE 2026.

Paper-I: Tue 29 September 2026, 1:00–4:00 pm · **Paper-II:** Wed 30 September, 9 am–12 · **MCQ:** Wed 30 September, 1–3 pm.

This file covers **Paper-I §8** (Computer languages, Software classification, and MS Office in detail) and **Paper-II §1** (the MS-DOS and Windows portion).

0. Why this is your highest-risk file (read first)

You are a **B.Tech Computer Science graduate**. You already know operating systems, DBMS, networks, C and data structures *above* the depth this Diploma paper asks — for those units your job is refresh, not learning. **This file is different.** As [CSDip_00_OVERVIEW.md](#) warns, MS Office and MS-DOS are computer-*literacy* content, not computer-*science* content. You very plausibly **never formally studied**:

- The **menu-level, feature-level** detail of Word, Excel, PowerPoint and Access — DropCap, WordArt, callouts, footnote-vs-endnote, tabs/borders/shading, **Mail Merge steps**, Excel function *categories*, chart components, PowerPoint *views*, Access *query types*.
- **MS-DOS** — the boot sequence, the system files, the exact **internal-vs-external** command split with syntax, and the meaning of **.BAT / .EXE / .COM**.

None of this is derivable from first principles. You cannot reason your way to "which DOS command is external" or "how many steps Mail Merge has." You have to *learn it as fact*. And it is **disproportionately represented in MCQs**, because each feature is an easy single-fact question. [CSDip_00](#) allocates **~25–30% of your total study time** to this file for maybe 15% of the syllabus. That ratio is correct. This is where the marks you can *lose* live.

⚠ **Version note.** Menu paths and some feature names differ slightly across Office versions (2003 menus vs 2007+ Ribbon; "Text" vs "Short Text" in Access). Where a detail is version-dependent it is marked ⚠ **verify**. Concepts (what Mail Merge *is*, what an absolute reference *does*) are version-independent and are given plainly. In the exam, **describe the concept and the standard steps**; do not stake an answer on an exact menu label unless you are sure.

Tick-box checklist — what "ready" means for this file

Tick a box only when you can write a 5-mark answer from memory *and* score 8/10 on a self-quiz of its facts.

Languages & software (Paper-I §7, §8a)

- ☐ Machine / assembly / high-level languages; 1GL → 5GL
- ☐ Characteristics of a good language
- ☐ Software classification: system vs application
- ☐ **Compiler vs Interpreter vs Assembler** (the comparison table)

MS Word (Paper-I §8b)

- ☐ Window components; editing; formatting
- ☐ Symbols / WordArt / DropCap; tabs, borders, shading
- ☐ Header/footer; **footnote vs endnote**

- [] Shapes, text boxes, callouts, captions; tables
- [] AutoCorrect; spelling & grammar; multiple columns
- [] ★ **Mail Merge — full steps in order (guaranteed 10-marker)**
- [] Shortcut keys table

MS Excel (Paper-I §8c)

- [] Window; workbook vs worksheet
- [] **Relative vs absolute vs mixed referencing** (worked)
- [] Function syntax: SUM, AVERAGE, COUNT, COUNTA, IF, nested IF, VLOOKUP, HLOOKUP, MAX, MIN, ROUND, CONCATENATE
- [] Sort / filter
- [] **Charts: components, types, 3-D**

MS PowerPoint (Paper-I §8d)

- [] Components; **the views and their uses**
- [] Templates / slide master
- [] **Transition vs animation**; action buttons

MS Access (Paper-I §8e)

- [] Tables / fields / records / data types; primary keys
- [] Relationships (1:1, 1:M, M:N)
- [] **Queries (select / parameter / action / QBE)**; forms; reports

MS-DOS (Paper-II §1f)

- [] Features; booting; system files (IO.SYS, MSDOS.SYS, COMMAND.COM)
- [] **Internal vs external commands — full categorised table with syntax**
- [] Wildcards; **BAT vs EXE vs COM**

Windows (Paper-II §1g)

- [] Window operations; file/folder management; Recycle Bin/Restore
- [] Accessories (Notepad vs WordPad, Paint); versions in order

PART A — COMPUTER LANGUAGES & SOFTWARE CLASSIFICATION

A.1 Computer languages — the generations

Intuition first. A computer only understands 0s and 1s. Every "language" above that is a more human-friendly layer that must eventually be **translated down** to those 0s and 1s. The higher the generation, the closer to human language and the further from the machine.

Gen	Name	What it is	Example	Needs translation?
1GL	Machine language	Pure binary (0s/1s); the only thing the CPU runs directly	10110000 01100001	No (it <i>is</i> machine code)
2GL	Assembly language	Mnemonics for machine instructions (ADD, MOV, SUB)	MOV AL, 61h	Yes — by an assembler
3GL	High-level language	English-like, machine-independent, procedural	C, FORTRAN, COBOL, BASIC, Pascal	Yes — compiler/interpreter
4GL	Very-high-level / problem-oriented	Closer to natural language; specify <i>what</i> , not <i>how</i>	SQL, report generators	Yes
5GL	AI / constraint-based	Solve problems via constraints and logic	Prolog, LISP (AI context)	Yes

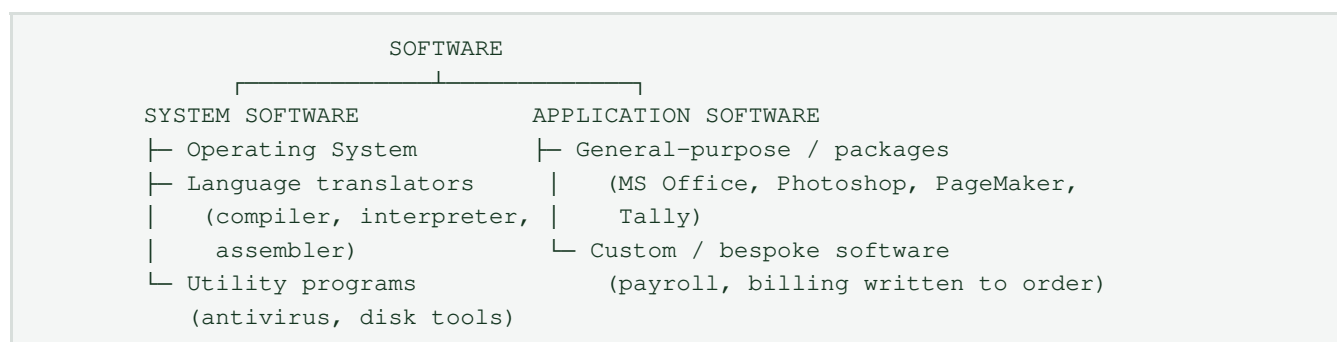
Machine vs Assembly vs High-level — the classic table:

Feature	Machine (1GL)	Assembly (2GL)	High-level (3GL)
Form	Binary 0/1	Mnemonics	English-like statements
Machine dependent?	Yes	Yes (CPU-specific)	No (portable)
Ease for humans	Hardest	Moderate	Easiest
Execution speed	Fastest	Fast	Slower (needs translation)
Translator needed	None	Assembler	Compiler / Interpreter

Characteristics of a good programming language (a common 5-marker): simplicity & readability, naturalness for its problem area, portability (machine independence), efficiency, reliability, ease of maintenance, good structure/modularity, availability of compilers and good documentation.

A.2 Software classification

Intuition. Software is the set of instructions that makes hardware useful. It splits into software that runs *the computer itself* (system) and software that does *your work* (application).



Type	Purpose	Examples
System software	Runs and manages the computer; a platform for applications	OS (Windows, Linux, DOS), compilers, interpreters, assemblers, device drivers, utilities
Application software	Helps the user do specific tasks	MS Word, MS Excel, Photoshop, PageMaker, Tally, browsers

A.3 ★ Compiler vs Interpreter vs Assembler (guaranteed question)

Intuition. All three are **translators** that convert human-written code into machine code. The difference is *what* they translate and *how* they do it.

Feature	Compiler	Interpreter	Assembler
Translates	High-level language → machine code	High-level language → machine code	Assembly language → machine code
How	Whole program at once	Line by line	Whole program (assembly)
Output	A separate object/executable file	No separate file; executes directly	Object code

Speed of execution	Faster (translated once)	Slower (translates each run)	Fast
Error reporting	Lists all errors after compiling	Stops at the first error	Reports errors in assembly
Memory	Needs space for object code	Less (no object file)	—
Examples	C, C++ compilers	Python, BASIC, older interpreted langs	MASM, TASM

MCQ traps:

- A **compiler translates the whole program at once**; an **interpreter one line at a time**.
- The **assembler** translates **assembly** language (not high-level).
- An interpreter reports **only the first error** and then stops; a compiler reports the full error list.

Part A — likely exam questions

- [5] Differentiate between a compiler and an interpreter.
- [5] Classify computer languages into generations with examples.
- [5] Distinguish between system software and application software.
- [10] Explain machine, assembly and high-level languages with their advantages and disadvantages.

PART B — MS WORD

B.1 The Word window — components

Intuition. Everything you do in Word happens in a window with a fixed furniture of bars and areas. Naming them is a common objective/5-mark item.

Component	What it is
Title bar	Top bar showing the document name and the program name
Quick Access Toolbar	Small customisable toolbar (Save, Undo, Redo)  verify (2007+)
Ribbon / Menu bar	The tabbed command area (Home, Insert, Page Layout, References, Mailings, Review, View) in 2007+; a menu bar in 2003  verify
Groups	Sections within a Ribbon tab (e.g. Font, Paragraph)
Ruler	Horizontal & vertical rulers for margins, tabs, indents
Document/Text area	The white page where you type
Cursor / Insertion point	The blinking vertical line where text appears
Scroll bars	Move the view vertically/horizontally
Status bar	Bottom bar: page number, word count, language, view buttons, zoom
View buttons	Print Layout, Web Layout, Read Mode, Outline, Draft

B.2 Editing & formatting

- **Editing:** typing, selecting (click-drag; double-click = word; triple-click = paragraph Δ verify), **Cut / Copy / Paste**, **Undo/Redo**, **Find & Replace**, Go To, insert/overtyping.
- **Character formatting (Font group):** font face, size, **bold**, *italic*, underline, colour, highlight, subscript/superscript, change case, strikethrough.
- **Paragraph formatting:** alignment (left/centre/right/justify), line spacing, indentation, bullets & numbering, borders & shading.
- **Page formatting:** margins, orientation (portrait/landscape), paper size, columns, page breaks, watermark, page colour, page borders.

B.3 Symbols, WordArt, DropCap

Feature	What it does	Where Δ verify menu
Symbol	Insert characters not on the keyboard (@, ±, §, α)	Insert → Symbol
WordArt	Decorative, stylised text with effects (curved, 3-D, coloured)	Insert → WordArt
Drop Cap	Enlarges the first letter of a paragraph to drop across several lines (as in a storybook)	Insert → Drop Cap; options Dropped or In Margin
Equation	Insert mathematical equations	Insert → Equation

B.4 Tabs, borders and shading

- **Tabs / Tab stops:** positions the cursor jumps to when you press **Tab**; types — **Left, Right, Centre, Decimal, Bar**; set via the ruler or Tabs dialog. Useful for aligning columns of text without a table.
- **Borders:** lines around text, paragraphs, pages, or table cells (style, colour, width). **Shading:** a background fill/colour behind text or a cell.

B.5 Header, footer, footnote and endnote

- **Header** = repeating text at the **top** of every page; **Footer** = repeating text at the **bottom** (page numbers, date, title, logo).

★ **Footnote vs Endnote (a guaranteed "differentiate" question):**

Feature	Footnote	Endnote
Appears	At the bottom of the same page where the reference mark is	At the end of the document (or section)
Purpose	A note/citation tied to a specific point	Consolidated notes/references at the end
Default numbering	1, 2, 3 ...	i, ii, iii ... Δ verify default
Inserted via	References → Insert Footnote	References → Insert Endnote
Shortcut Δ verify	Alt+Ctrl+F	Alt+Ctrl+D

B.6 Shapes, text boxes, callouts, captions, tables

Feature	What it is
Shapes	Drawing objects — lines, rectangles, circles, arrows, stars, flowchart symbols
Text box	A movable box that holds text anywhere on the page, independent of the main flow
Callout	A shape with a pointer/leader line used to label or annotate (like a speech bubble)
Caption	A numbered label added below/above a figure or table ("Figure 1: ...", "Table 2: ...")
Table	Rows × columns grid; insert, merge/split cells, borders, sort, formulas (=SUM) △ verify

B.7 AutoCorrect, spelling & grammar, multiple columns

- **AutoCorrect:** automatically fixes common typos ("teh" → "the") and can expand abbreviations as you type.
- **Spelling & Grammar check:** red wavy underline = spelling, green/blue wavy = grammar; run with **F7** △ verify.
- **Multiple columns:** split page text into 2 or more newspaper-style columns (Layout → Columns): One, Two, Three, Left, Right, or custom width/spacing with a line between.

B.8 ★ MAIL MERGE — the guaranteed 10-marker (learn the steps in order)

Intuition. Mail Merge lets you write **one** letter and automatically produce **many** personalised copies — same body, different name/address on each — by combining a **main document** with a **data source (recipient list)**. Classic use: sending the same letter to 500 people, each addressed personally.

The three ingredients:

Ingredient	What it is
Main document	The letter/email/label template with fixed text
Data source	The list of recipients (a Word table, Excel sheet, Access DB, or Outlook contacts) with fields like Name, Address, City
Merge fields	Placeholders in the main document («Name», «Address») that get replaced by data-source values

The steps (Mailings tab → Start Mail Merge → Step-by-Step Wizard):

1. **Select the document type** — Letters, E-mail messages, Envelopes, Labels, or Directory. (Usually **Letters**.)
2. **Select the starting document** — the current document, a template, or an existing one.
3. **Select recipients** — choose the data source: **type a new list, use an existing list** (Excel/Access/Word table), or **select from Outlook contacts**.
4. **Write / arrange your document** — type the fixed text and **insert merge fields** at the right places: **Address Block**, **Greeting Line**, or **Insert Merge Field** for individual fields.
5. **Preview your letters** — check how each personalised copy looks; scroll through

recipients; exclude any you don't want.

6. **Complete the merge** — **Print** the letters directly, or **Edit individual documents** to produce one merged file you can review/save.

Answer tip: write the definition, the **three ingredients**, then the **six numbered steps**. Drawing the "main document + data source → merged letters" flow scores well.

B.9 Word shortcut keys (memorise — MCQ gold)

Shortcut	Action	Shortcut	Action
Ctrl+N	New document	Ctrl+B	Bold
Ctrl+O	Open	Ctrl+I	Italic
Ctrl+S	Save	Ctrl+U	Underline
Ctrl+P	Print	Ctrl+E	Centre align
Ctrl+C	Copy	Ctrl+L	Left align
Ctrl+X	Cut	Ctrl+R	Right align
Ctrl+V	Paste	Ctrl+J	Justify
Ctrl+Z	Undo	Ctrl+A	Select all
Ctrl+Y	Redo	Ctrl+F	Find
Ctrl+K	Insert hyperlink	Ctrl+H	Replace
F7	Spelling & grammar	Ctrl+Home / Ctrl+End	Start / end of document

Part B (Word) — likely exam questions

- **[10]** What is Mail Merge? Explain the steps to perform a mail merge in MS Word.
- **[5]** Differentiate between a footnote and an endnote.
- **[5]** What is Drop Cap? What is WordArt?
- **[5]** Explain header and footer.
- **[10]** Describe the main components of the MS Word window.
- **[5]** Differentiate between a text box and a callout.

Part B — MCQ traps (with answers)

Answers

1. A footnote appears at the? → ****bottom of the same page**** 2. Mail Merge combines a main document with a? → ****data source (recipient list)**** 3. Ctrl+U does? → ****Underline**** 4. Drop Cap affects the? → ****first letter of a paragraph**** 5. F7 is the shortcut for? → ****Spelling & Grammar check**** 6. Repeating text at the top of every page is a? → ****Header**** 7. Which tab contains Mail Merge? → ****Mailings**** △ verify (2007+)

PART C — MS EXCEL

C.1 The Excel window & basic terms

Intuition. Excel is a grid of cells for storing numbers, text and **formulas** that recalculate automatically when data changes.

Term	Meaning
Cell	Intersection of a column and row (e.g. B3)
Cell reference / address	Column letter + row number (B3)
Column	Vertical; labelled A, B, C ...
Row	Horizontal; labelled 1, 2, 3 ...
Name box	Shows the active cell's address; top-left
Formula bar	Where you type/edit a cell's content or formula
Active cell	The currently selected cell (bordered)
Range	A block of cells, e.g. A1:A10 or A1:C5
Sheet tabs	Tabs at the bottom to switch worksheets

C.2 Workbook vs Worksheet (a "differentiate" question)

Feature	Workbook	Worksheet
What it is	The whole Excel file	A single sheet (page) inside the file
Contains	One or more worksheets	Cells (rows & columns)
File extension	.xlsx (2007+) / .xls (2003) △ verify	— (it lives inside the workbook)
Analogy	A book	A page in the book

C.3 ★ Relative, absolute and mixed referencing (worked — high yield)

Intuition. When you **copy a formula** to another cell, Excel adjusts the cell references *unless you lock them*. How they change is the single most-tested Excel concept. The \$ sign locks a part; press **F4** to cycle through the forms.

Type	Looks like	Behaviour when copied	Use it when...
Relative	A1	Both column and row adjust	you want the reference to move with the formula (the normal case)
Absolute	\$A\$1	Neither changes — fully locked	you always point to one fixed cell (e.g. a tax rate)
Mixed	\$A1 (column locked) or A\$1 (row locked)	only the unlocked part adjusts	you lock one direction, e.g. multiplication tables

Worked example. Cell **C1** contains **=A1*B1**. Copy it **down** to C2:

	If C1 is	C2 becomes	Why
Relative	=A1*B1	=A2*B2	both rows shift down by one

Absolute	<code>=A\$1*\$B\$1</code>	<code>=A\$1*\$B\$1</code>	fully locked — no change
Mixed	<code>=A\$1*B1</code>	<code>=A\$1*B2</code>	row 1 in the first term is locked, the rest shifts

MCQ trap: `$A$1` is **absolute** (locked both ways); `$A1` and `A$1` are **mixed**; `A1` is **relative**. The `$` locks whatever comes **immediately after** it.

C.4 Function syntax — the must-know functions

Intuition. A function is a built-in formula. Syntax = `=FUNCTION(arguments)`.

Function	Syntax	Returns
SUM	<code>=SUM(A1:A10)</code>	Total of the range
AVERAGE	<code>=AVERAGE(A1:A10)</code>	Arithmetic mean
COUNT	<code>=COUNT(A1:A10)</code>	Count of cells that contain numbers
COUNTA	<code>=COUNTA(A1:A10)</code>	Count of non-empty cells (numbers and text)
MAX	<code>=MAX(A1:A10)</code>	Largest value
MIN	<code>=MIN(A1:A10)</code>	Smallest value
ROUND	<code>=ROUND(3.14159, 2) → 3.14</code>	Rounds a number to given decimals
IF	<code>=IF(A1>=40, "Pass", "Fail")</code>	One value if true, another if false
Nested IF	<code>=IF(A1>=90, "A", IF(A1>=75, "B", "C"))</code>	Tests multiple conditions in sequence
VLOOKUP	<code>=VLOOKUP(lookup_value, table, col_index, [range_lookup])</code>	Looks up a value in the first column of a table and returns a value from another column (vertical)
HLOOKUP	<code>=HLOOKUP(lookup_value, table, row_index, [range_lookup])</code>	Same but searches the first row (horizontal)
CONCATENATE	<code>=CONCATENATE(A1, " ", B1)</code>	Joins text from several cells into one

Key distinctions (MCQ traps):

- **COUNT** counts only **numbers**; **COUNTA** counts **all non-empty** cells.
- **VLOOKUP** searches the **first column** and looks **down** (Vertical); **HLOOKUP** searches the **first row** and looks **across** (Horizontal).
- Every formula/function **begins with =**.

C.5 Sort & Filter

- **Sort:** rearrange rows by a column's values — **Ascending (A→Z, small→large)** or **Descending (Z→A, large→small)**; can sort by multiple levels (e.g. by City, then Name).
- **Filter (AutoFilter):** temporarily **hides** rows that don't match a condition, showing only the rows you want; the data isn't deleted, just hidden.

C.6 Charts — components, types, 3-D

Intuition. A chart turns numbers into a picture so trends are visible at a glance.

Chart components (label these for a 5-marker):

Component	What it is
Chart area	The whole chart region
Plot area	The inner region where the data is drawn
Data series	A set of related values (one column/row of data)
Data point	A single value in a series
Axes	X-axis (category) and Y-axis (value)
Legend	Key that identifies each data series by colour
Gridlines	Reference lines across the plot area
Data labels	The actual values shown on the bars/points
Chart title / axis titles	Names of the chart and axes

Chart types (know when each is used):

Type	Best for
Column / Bar	Comparing values across categories (vertical/horizontal bars)
Line	Trends over time
Pie	Parts of a whole (percentages) — one data series
Area	Magnitude of change over time
XY (Scatter)	Relationship/correlation between two variables
Doughnut	Like a pie but multiple series
Radar / Stock / Surface / Bubble	Specialised comparisons △ verify list per version

3-D charts: many chart types (column, bar, pie, line, area) have a **three-dimensional variant** that adds depth for visual effect — e.g. a **3-D pie** or **3-D column**. They look striking but can distort perception of values, so they are used for presentation impact.

Part C (Excel) — likely exam questions

- **[10]** Explain relative, absolute and mixed cell referencing with examples.
- **[5]** Differentiate between COUNT and COUNTA.
- **[5]** What is the difference between VLOOKUP and HLOOKUP?
- **[10]** Explain the components of a chart. Name any four chart types and their uses.
- **[5]** Differentiate between a workbook and a worksheet.
- **[5]** Write the syntax of the IF function with an example of a nested IF.

Part C — MCQ traps (with answers)

Answers

1. `A\$1` is which type of reference? → **Absolute** 2. `A\$1` is which type? → **Mixed** (row locked) 3. COUNT counts cells containing? → **numbers only** 4. VLOOKUP searches which part of the table? → **the**

first (leftmost) column** 5. Every Excel formula starts with? → **= (equal sign)** 6. A pie chart shows? → **parts of a whole (one data series)** 7. The whole Excel file is called a? → **Workbook** 8. Which key cycles reference types? → **F4** △ verify

PART D — MS POWERPOINT

D.1 Components

Intuition. PowerPoint builds a **slide show** — a sequence of slides holding text, images, charts and media for a presentation.

Component	What it is
Slide	A single page of the presentation
Slide pane	The main area where you edit the current slide
Slides/Outline tab (thumbnails)	Left-side panel showing all slides in order
Placeholder	A pre-formatted box for title, text, or content
Notes pane	Space for the speaker's private notes
Status bar	Slide number, view buttons, zoom

D.2 ★ The views and their uses (guaranteed question)

View	What it shows / used for
Normal view	Default working view — edit one slide at a time (slide + thumbnails + notes)
Slide Sorter view	All slides as thumbnails — reorder, add, delete, duplicate slides and apply transitions
Notes Page view	The slide plus a large area for speaker notes , as it will print
Reading view	Play the show in a window (not full screen) for review
Slide Show view	The full-screen presentation the audience sees (start with F5) △ verify
Outline view	Shows only the text of slides as an outline — good for organising content quickly △ verify

D.3 Templates & Slide Master

- **Design template / theme:** a ready-made set of colours, fonts and background applied to all slides for a consistent look.
- **Slide Master:** the **top-level slide** that controls the formatting of **every** slide — change the master once and all slides update (fonts, logo, background, placeholders). This is a favourite exam concept: "a change made in the Slide Master applies to all slides."

D.4 ★ Transition vs Animation (guaranteed "differentiate")

Intuition. A **transition** is the effect **between slides** (how one slide gives way

to the next). An **animation** is the effect applied to **objects within a slide** (how a bullet, image or title enters, exits or moves).

Feature	Transition	Animation
Applies to	The whole slide (slide-to-slide change)	An individual object on a slide (text, image, shape)
Example	Fade, Push, Wipe, Cover between slides	Fly In, Appear, Wipe, Zoom of a bullet point
Purpose	Smooth movement from one slide to the next	Control the order/emphasis of elements on one slide
Ribbon tab  verify	Transitions	Animations

D.5 Action buttons

Action buttons are ready-made shapes (Home, Next, Back, Forward, Return) you place on a slide; clicking them during the show performs an **action** — jump to another slide, open a file/URL, or run a program. Used to make presentations interactive/navigable.

Part D (PowerPoint) — likely exam questions

- [5] Explain the different views in MS PowerPoint and their uses.
- [5] Differentiate between a slide transition and an animation.
- [5] What is a Slide Master? Why is it useful?
- [5] What are action buttons? Give two examples.

Part D — MCQ traps (with answers)

Answers

1. Which view is used to reorder slides? → **Slide Sorter view** 2. An effect between two slides is a? → **Transition** 3. An effect applied to a bullet point is an? → **Animation** 4. Changing the Slide Master affects? → **all slides** 5. F5 starts? → **the Slide Show**  verify 6. Speaker notes are added in? → **Notes pane / Notes Page view**

PART E — MS ACCESS

E.1 The building blocks

Intuition. Access is a **relational database** program. Instead of one flat sheet, data lives in **tables** that can be **linked**, and you interrogate them with **queries**, enter data through **forms**, and print results as **reports**.

Term	Meaning
Database	An organised collection of related data
Table	A grid storing data about one subject (e.g. Students)
Field	A column — one attribute (e.g. Name, RollNo)

Record	A row — all data about one entity (one student)
Data type	The kind of value a field holds
Primary key	A field that uniquely identifies each record; no duplicates, no nulls

E.2 Field data types

Data type	Holds	Example
Short Text / Text	Text up to ~255 chars $\triangle$ verify	Name, address
Long Text / Memo	Long text/paragraphs	Remarks
Number	Numeric values for calculation	Age, marks
Date/Time	Dates and times	DOB
Currency	Money values	Fees
AutoNumber	Auto-incrementing unique number (often a primary key)	ID
Yes/No (Boolean)	Two-state true/false	Passed?
OLE Object	Embedded object (image, file)	Photo
Hyperlink	Web/email link	Website
Attachment $\triangle$ verify	Attached files	—

E.3 The Access objects

Object	Purpose
Table	Stores the actual data
Query	Retrieves/filters/updates data (see E.5)
Form	A user-friendly screen for entering and viewing records one at a time
Report	A formatted printout/summary of data (with grouping, totals)
Macro / Module	Automation $\triangle$ verify (beyond Diploma depth)

E.4 Relationships

Intuition. Tables are linked through a **common field** (a primary key in one table matching a **foreign key** in another). This avoids duplicating data.

Relationship	Meaning	Example
One-to-One (1:1)	One record in A links to one in B	Person $\leftrightarrow$ Passport
One-to-Many (1:M)	One record in A links to many in B (most common)	Department $\leftrightarrow$ Employees
Many-to-Many (M:N)	Many in A link to many in B (needs a junction table)	Students $\leftrightarrow$ Courses

E.5 ★ Queries — the types (guaranteed question)

Intuition. A query asks a question of the data ("show all students who scored > 60").

Query type	What it does
Select query	Retrieves and displays records matching criteria (the most common)
Parameter query	Prompts the user to type a value at run-time (e.g. "Enter city:") and uses it as the criterion
Action query	Changes data in bulk — four kinds: Append (add records), Update (modify), Delete (remove), Make-Table (create a new table from results)
Crosstab query	Summarises data in a spreadsheet-like grid (row × column totals) △ verify
QBE (Query By Example)	The design-grid method of building a query visually — you specify fields and criteria in a grid instead of writing SQL

MCQ trap: QBE = Query By Example, the visual grid way of designing a query. The four **action** queries are **Append, Update, Delete, Make-Table**.

Part E (Access) — likely exam questions

- [15] Explain MS Access — database planning, tables and data types, queries, forms and reports.
- [5] Differentiate between a field and a record.
- [5] What is a primary key? Why is it important?
- [5] Explain the types of queries in MS Access.
- [5] Explain the three types of relationships in a database.

Part E — MCQ traps (with answers)

Answers

1. A column in a table is a? → **Field** 2. A row in a table is a? → **Record** 3. A field that uniquely identifies each record? → **Primary key** 4. QBE stands for? → **Query By Example** 5. The four action queries? → **Append, Update, Delete, Make-Table** 6. Most common relationship type? → **One-to-Many (1:M)** 7. A user-friendly data-entry screen is a? → **Form**

PART F — MS-DOS

F.1 Features & characteristics

Intuition. MS-DOS (Microsoft Disk Operating System) is an old, **command-line, single-user, single-tasking** operating system — you type commands rather than click icons. Understanding it is pure recall; nothing is derivable.

Feature	Detail
Interface	Command-line (CLI) — you type commands at a prompt (C:\>)
User/task model	Single-user, single-tasking
Character-based	No graphics (text only)

Case-insensitive	<code>DIR = dir</code>
File naming	8.3 format — up to 8 characters name + 3 characters extension (e.g. <code>LETTER.DOC</code>)
16-bit	Runs 16-bit programs

F.2 The booting process

Intuition. "Booting" = starting the computer and loading the OS into memory. (The word comes from "pull yourself up by your bootstraps.")

- **Cold boot** = starting from power-off. **Warm boot** = restarting a running machine (**Ctrl+Alt+Del**) without cutting power.

Boot sequence (learn the order):

1. **Power on** → the CPU runs the **BIOS**.
2. **POST (Power-On Self-Test)** — BIOS checks the hardware (RAM, keyboard, disks).
3. BIOS loads the **boot loader** from the **boot sector** of the disk (**MBR**).
4. **IO.SYS** loads (handles basic input/output).
5. **MSDOS.SYS** loads (the kernel/core of DOS).
6. **CONFIG.SYS** is read (loads device drivers & settings) — *if present*.
7. **COMMAND.COM** loads (the **command interpreter** / shell).
8. **AUTOEXEC.BAT** runs (a batch file of startup commands) — *if present*.
9. The **DOS prompt** `C:\>` appears — ready for commands.

F.3 System files (memorise these three + two config files)

File	Role	Type
IO.SYS	Handles low-level input/output; interfaces with BIOS	Hidden system file
MSDOS.SYS	The kernel — core of the OS; manages files, memory, programs	Hidden system file
COMMAND.COM	The command interpreter (shell) — reads and executes your typed commands; holds all internal commands	Command processor
<code>CONFIG.SYS</code>	Loads device drivers and system settings at boot	Text config file
<code>AUTOEXEC.BAT</code>	Batch file of commands run automatically at startup	Batch file

The three "system files" the syllabus names are **IO.SYS**, **MSDOS.SYS** and **COMMAND.COM**. `CONFIG.SYS` and `AUTOEXEC.BAT` are *startup configuration* files, not the core system files — but know both sets.

F.4 ★ Internal vs External commands (the key table with syntax)

Intuition — the single most important DOS distinction:

- **Internal commands** are **built into COMMAND.COM** — they load into memory at boot, so

they are **always available** and need no separate file on disk.

- **External commands** are **separate program files** (`.COM` / `.EXE`) stored on disk — DOS must **find and load the file** to run them, so they only work if that file (or its PATH) is present.

INTERNAL COMMANDS (in `COMMAND.COM`):

Command	Syntax	Purpose
DIR	<code>DIR [drive:][path] [/P] [/W]</code>	List files & folders in a directory
CD / CHDIR	<code>CD [path]</code>	Change the current directory
MD / MKDIR	<code>MD dirname</code>	Make (create) a directory
RD / RMDIR	<code>RD dirname</code>	Remove an (empty) directory
COPY	<code>COPY source destination</code>	Copy one or more files
DEL / ERASE	<code>DEL filename</code>	Delete file(s)
REN / RENAME	<code>REN oldname newname</code>	Rename a file
TYPE	<code>TYPE filename</code>	Display a text file's contents on screen
CLS	<code>CLS</code>	Clear the screen
DATE	<code>DATE</code>	Display/set the system date
TIME	<code>TIME</code>	Display/set the system time
VER	<code>VER</code>	Show the DOS version
PROMPT	<code>PROMPT \$p\$g</code>	Change the look of the command prompt
PATH	<code>PATH=C:\DOS;...</code>	Set the search path for executable files

EXTERNAL COMMANDS (separate `.EXE/.COM` files on disk):

Command	Syntax	Purpose
FORMAT	<code>FORMAT drive:</code>	Prepare (format) a disk for use — erases it
DISKCOPY	<code>DISKCOPY A: B:</code>	Make an exact copy of one floppy to another
CHKDSK	<code>CHKDSK drive:</code>	Check a disk for errors and report status
SCANDISK	<code>SCANDISK drive:</code>	Scan and repair disk errors ⚠ verify (later DOS)
XCOPY	<code>XCOPY source dest /S</code>	Copy files and subdirectories (more powerful COPY)
ATTRIB	<code>ATTRIB +R file</code>	View/change file attributes (Read-only, Hidden, System, Archive)
TREE	<code>TREE [drive:]</code>	Display the directory structure graphically
LABEL	<code>LABEL drive: name</code>	Create/change a disk's volume label
SORT	<code>SORT < file</code>	Sort lines of text
FIND	<code>FIND "text" file</code>	Search a file for a text string

MORE	<code>MORE < file</code>	Display output one screen at a time
EDIT	<code>EDIT filename</code>	Open the full-screen text editor

How to tell them apart in the exam: the "everyday navigation and file" commands (DIR, CD, MD, RD, COPY, DEL, REN, TYPE, CLS, DATE, TIME, VER, PROMPT, PATH) are **internal**. The "disk utilities and tools" (FORMAT, DISKCOPY, CHKDSK, SCANDISK, XCOPY, ATTRIB, TREE, LABEL, SORT, FIND, MORE, EDIT) are **external**. Memorise the two lists; this is a guaranteed "differentiate + list" question and a rich MCQ source.

F.5 Wildcards

Wildcards let one command act on many files whose names match a pattern:

Wildcard	Meaning	Example
* (asterisk)	Matches any number of characters	<code>DIR *.TXT</code> = all files with <code>.TXT</code> extension; <code>DEL *.*</code> = all files
? (question mark)	Matches exactly one character	<code>DIR FILE?.DOC</code> matches FILE1.DOC, FILEA.DOC, but not FILE12.DOC

F.6 ★ BAT vs EXE vs COM (guaranteed "differentiate")

Intuition. All three can be **run by typing their name** at the prompt, but they are fundamentally different kinds of file.

Feature	.COM	.EXE	.BAT
Type	Executable — a direct memory image of a program	Executable — with a header, relocatable	Batch file — plain text list of DOS commands
Size limit	Max 64 KB (single segment)	Can be large (multi-segment)	Any (it's just text)
How it runs	Loaded directly and executed	Loaded via its header, then executed	Interpreted line by line by COMMAND.COM
Human-readable?	No (binary)	No (binary)	Yes (you can open/edit it in a text editor)
Example	<code>COMMAND.COM</code>	<code>FORMAT.EXE</code>	<code>AUTOEXEC.BAT</code>

Execution priority — if `RUN.COM`, `RUN.EXE` and `RUN.BAT` all exist and you type `RUN`, DOS runs them in the order `.COM` → `.EXE` → `.BAT`. (COM first.)

MCQ traps:

- **.COM** files have a **64 KB** size limit; **.EXE** can be larger.
- **.BAT** is a **text** file interpreted line by line — the only human-readable one.
- Priority order **COM > EXE > BAT**.

Part F (DOS) — likely exam questions

- **[10]** Differentiate between internal and external DOS commands, with examples of each.

- [5] Explain the booting process of MS-DOS.
- [5] Name the system files of MS-DOS and state the role of each.
- [5] Differentiate between .BAT, .EXE and .COM files.
- [5] What are wildcards? Explain * and ? with examples.
- [10] Explain any five internal and any five external DOS commands with syntax.

Part F — MCQ traps (with answers)

Answers

1. Internal commands are stored in? → **COMMAND.COM** 2. FORMAT is an? → **external** command 3. DIR is an? → **internal** command 4. The command interpreter of DOS is? → **COMMAND.COM** 5. * wildcard matches? → **any number of characters** 6. Max size of a .COM file? → **64 KB** 7. DOS 8.3 naming means? → **8-char name + 3-char extension** 8. Warm boot key combination? → **Ctrl+Alt+Del** 9. Which loads first: IO.SYS or COMMAND.COM? → **IO.SYS** 10. The DOS kernel file is? → **MSDOS.SYS**

PART G — WINDOWS OS

G.1 Window operations

Intuition. Windows is a **GUI (Graphical User Interface)**, **multi-tasking** OS — you interact with icons, windows and a mouse instead of typed commands.

Operation	How
Open / Close	Double-click an icon to open; click the ✕ to close
Minimize	Shrink the window to the taskbar (the _ button)
Maximize / Restore	Fill the screen (□) / return to previous size
Move	Drag the title bar
Resize	Drag a border or corner
Switch between windows	Alt+Tab , or click the taskbar

Desktop parts: icons, **Taskbar**, **Start button/menu**, **System tray (notification area)**, and the clock.

G.2 File & folder management

Task	How
Create folder	Right-click → New → Folder
Copy / Move	Copy (Ctrl+C) then Paste (Ctrl+V); Move = Cut (Ctrl+X) then Paste
Rename	Right-click → Rename, or press F2
Delete	Delete key → goes to Recycle Bin

Search	Search box / File Explorer search
Tool	File Explorer / Windows Explorer is the file-management program

G.3 Recycle Bin & Restore

Intuition. When you delete a file it isn't destroyed immediately — it goes to the **Recycle Bin**, a holding area, so you can recover it.

- **Restore:** open the Recycle Bin → right-click the file → **Restore** → it returns to its original location.
- **Permanent delete:** emptying the Recycle Bin, or **Shift+Delete** (bypasses the Bin).
- Files deleted from **removable drives** / **network drives** usually do **not** go to the Recycle Bin. △ verify

G.4 Accessories

Accessory	What it is	Key distinction
Notepad	A plain-text editor (.txt)	No formatting — no bold, fonts, colours
WordPad	A basic word processor	Supports formatting — fonts, bold, colour, images (.rtf , .doc)
Paint	A simple graphics/drawing program	Create/edit images (.bmp , .png , .jpg)
Calculator	On-screen calculator	Standard/scientific modes

★ **Notepad vs WordPad (common "differentiate"):** Notepad edits plain text with no formatting; WordPad is a lightweight word processor that supports fonts, styles, colours and images.

G.5 Windows versions in order (memorise the sequence)

Version	Year △ verify years
Windows 1.0	1985
Windows 3.0 / 3.1	1990 / 1992
Windows 95	1995
Windows 98	1998
Windows ME	2000
Windows 2000	2000
Windows XP	2001
Windows Vista	2007
Windows 7	2009

Windows 8 / 8.1	2012 / 2013
Windows 10	2015
Windows 11	2021

Learn the **order** (which came before which) even if you're unsure of exact years — "Which came first, Windows 98 or Windows XP?" is a typical MCQ. (98 came first.)

Part G (Windows) — likely exam questions

- [5] Differentiate between Notepad and WordPad.
- [5] What is the Recycle Bin? How do you restore a deleted file?
- [5] Explain file and folder management operations in Windows.
- [5] List the Windows versions in chronological order.

Part G — MCQ traps (with answers)

Answers

1. Deleted files go to the? → **Recycle Bin** 2. Which key permanently deletes (bypasses the Bin)? → **Shift+Delete** 3. Notepad saves files with which extension? → **.txt** 4. Which supports formatting: Notepad or WordPad? → **WordPad** 5. F2 is used to? → **Rename** 6. Alt+Tab does? → **switch between open windows** 7. Which came first: Windows 98 or Windows XP? → **Windows 98** 8. The file-management tool in Windows is? → **File Explorer / Windows Explorer**

Final revision map for this file

Priority	Block	Why it matters most
1	Mail Merge (B.8), Internal vs External DOS commands (F.4), BAT/EXE/COM (F.6)	Named guaranteed questions; pure recall you can't derive
2	Excel referencing (C.3) + functions (C.4), PowerPoint views (D.2) & transition-vs-animation (D.4)	High-frequency "differentiate/explain" items
3	Footnote vs endnote (B.5), Access query types (E.5), compiler/interpreter/assembler (A.3)	Classic 5-mark and MCQ fodder
4	DOS boot sequence (F.2) & system files (F.3), chart components (C.6)	Ordered-list and labelling questions
5	Windows: Notepad vs WordPad, Recycle Bin, versions (Part G)	Easy product-fact MCQs

One-line summary: This is the material you are least likely to already know and the material most likely to appear as easy MCQs and standard "differentiate/explain" descriptive questions. Over-learn **Mail Merge**, the **internal-vs-external DOS command split**, **BAT/EXE/COM**, **Excel referencing** and **PowerPoint views** — these are named, predictable, high-yield, and cannot be bluffed. Treat every **verify** menu label or year as "confirm before you write it," but the **concepts** in this file are exam-solid.